FPT UNIVERSITY HO CHI MINH CITY CAMPUS

DEPARTMENT OF SEMICONDUCTOR AND AUTOMOTIVE ENGINEERING



FPT UNIVERSITY

LAB REPORT

IC DESIGN INTRODUCTION

LAB 1: Study of MOSFET Characteristics

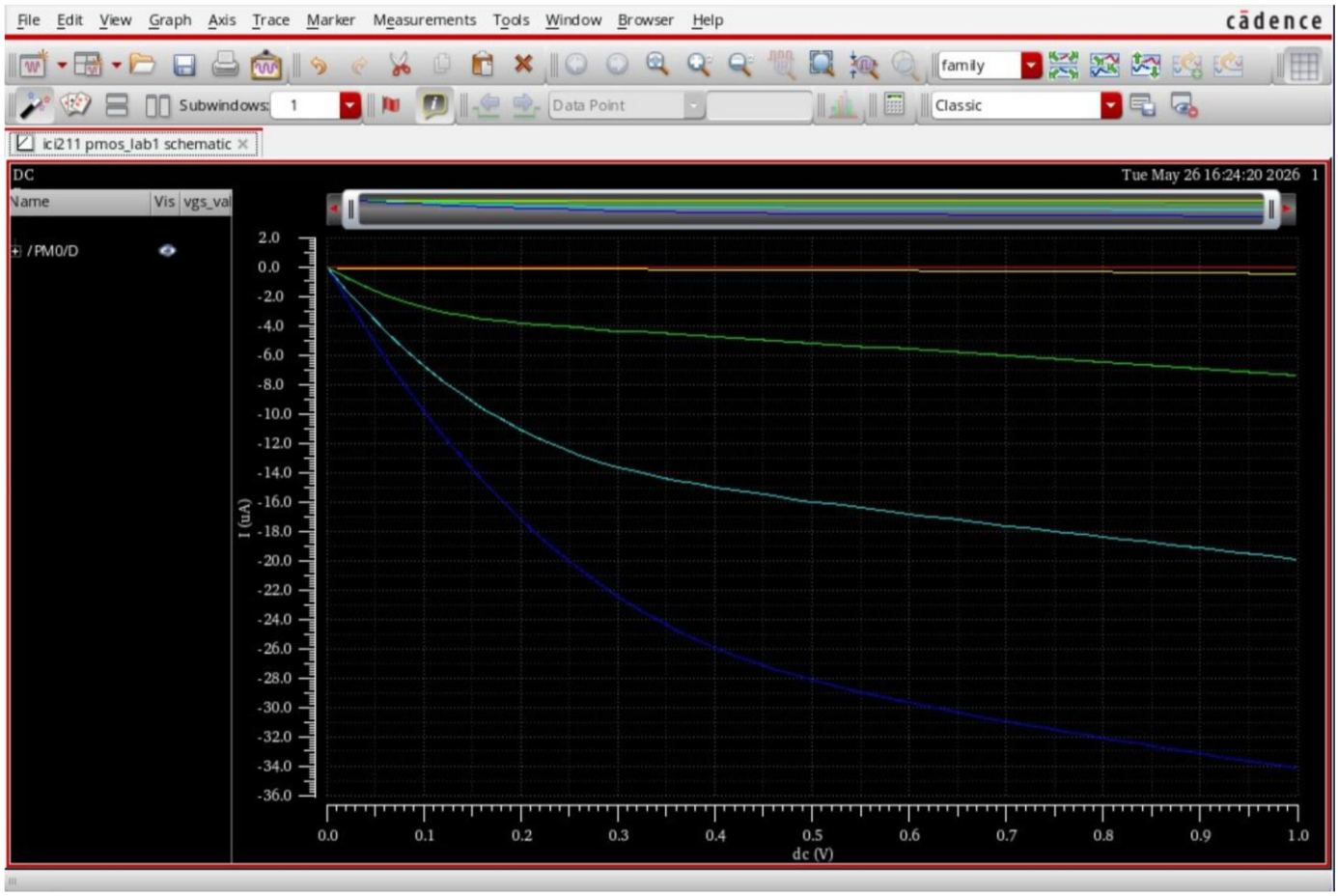
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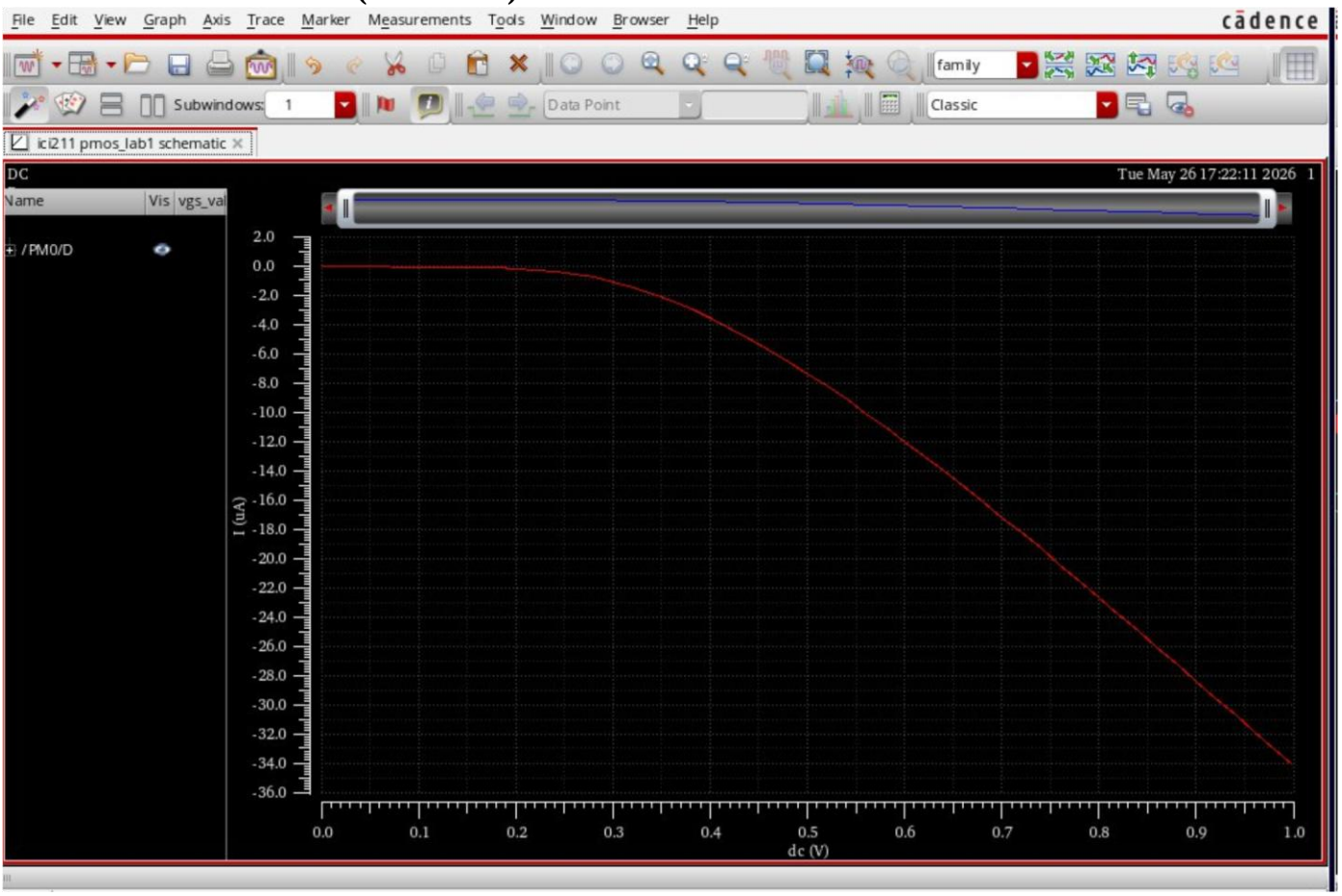
A.PMOS Characteristic Analysis

1.OutputCharacteristics(IDvsVDS)



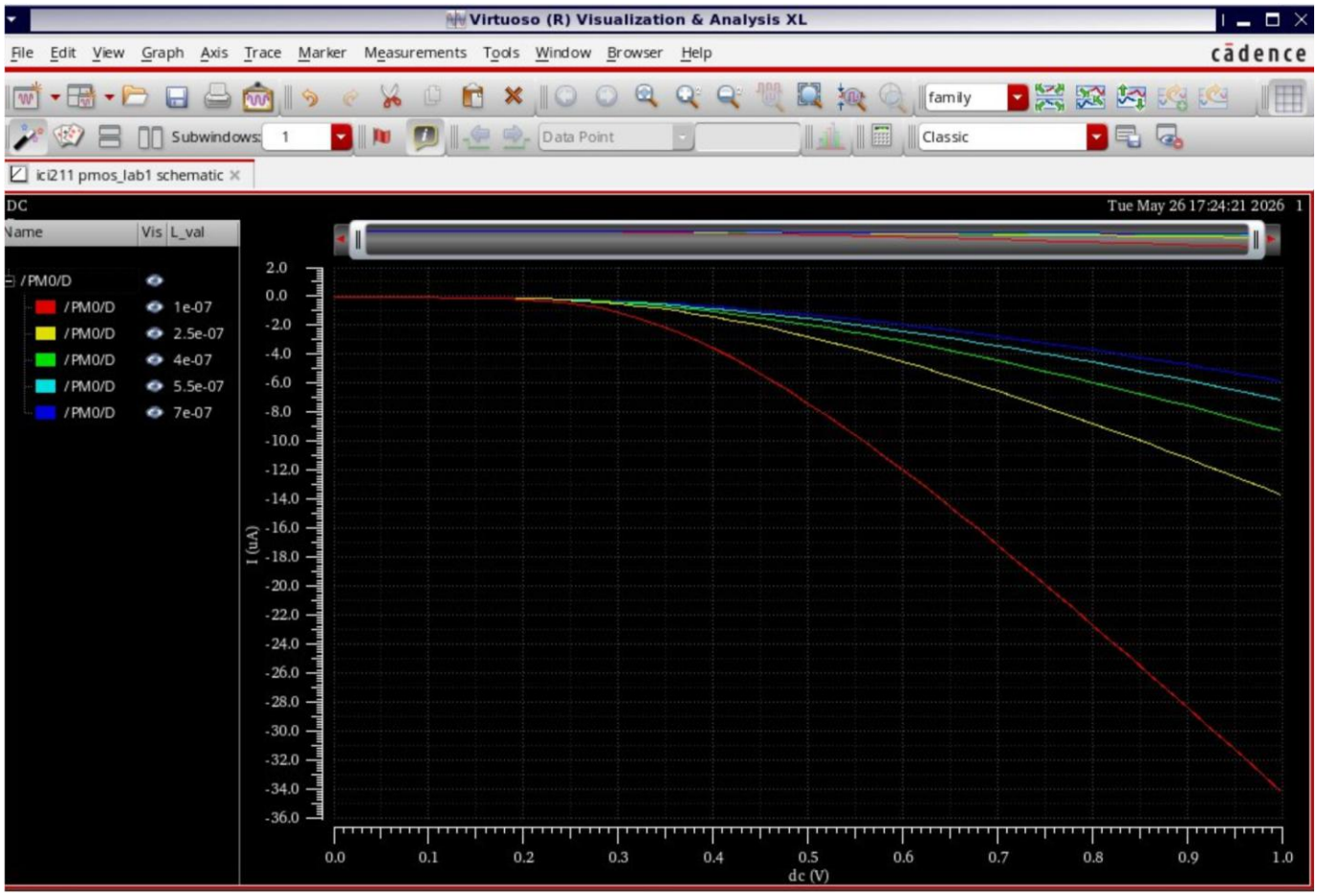
The output characteristic curves (I_D vs. V_{DS}) of the simulated PMOS device provide a clear visualization of its operational behavior under varying V_{GS} biases. As the gate-to-source potential intensifies, channel enhancement drives a corresponding surge in drain current. The characteristics capture the expected transition from the deep triode to the saturation regime. Furthermore, the non-ideal, finite slope observed during saturation accurately models Channel Length Modulation (CLM). Consequently, this confirms the finite nature of the transistor's output resistance, an essential metric that requires careful compensation in analog integrated circuit design.

2.Transfer Characteristics (ID vs VGS)



This simulation plot depicts the transfer characteristic of the PMOS transistor, illustrating the direct relationship between the drain current (I_D) and the gate-source voltage (V_{GS}). From 0V to approximately 0.3V, the device operates in the subthreshold region; here, the conductive channel has not yet formed, rendering I_D virtually zero. As the magnitude of the control voltage surpasses the threshold voltage (V_{th}, located between 0.3V and 0.4V), the transistor enters the strong inversion region. In this regime, the magnitude of I_D increases rapidly, following the established square-law model. A critical parameter derived from this characteristic curve is its slope, which defines the device's transconductance (g_m).

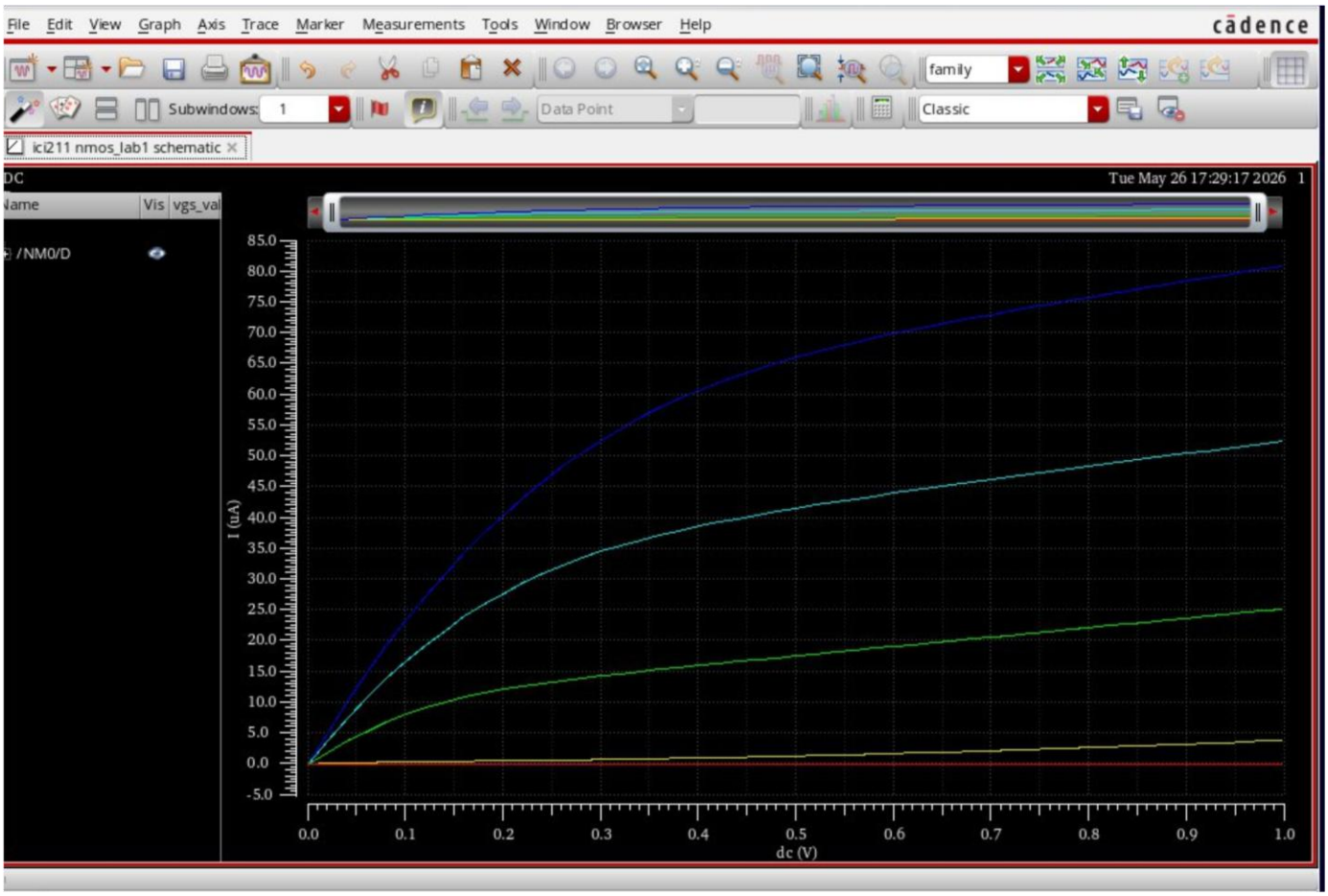
3.Effect of W/L Ratio:



This simulation graph highlights the geometric dependence of the PMOS drain current (I_D) on the channel length (L) through a parametric sweep. By systematically increasing L from 100 nm to 700 nm while holding the channel width (W) constant, the W/L aspect ratio decreases. This inversely affects the device, causing a proportional reduction in both the overall current magnitude and the curve's slope, which dictates the transconductance (g_m). Although minimumlength devices offer maximum current drive and g_m, analog IC designers often intentionally select larger channel lengths to suppress Channel Length Modulation (CLM). This deliberate design trade-off yields a higher intrinsic output resistance (r_o) and enhances both device matching and noise immunity—vital parameters for robust analog amplifier design.

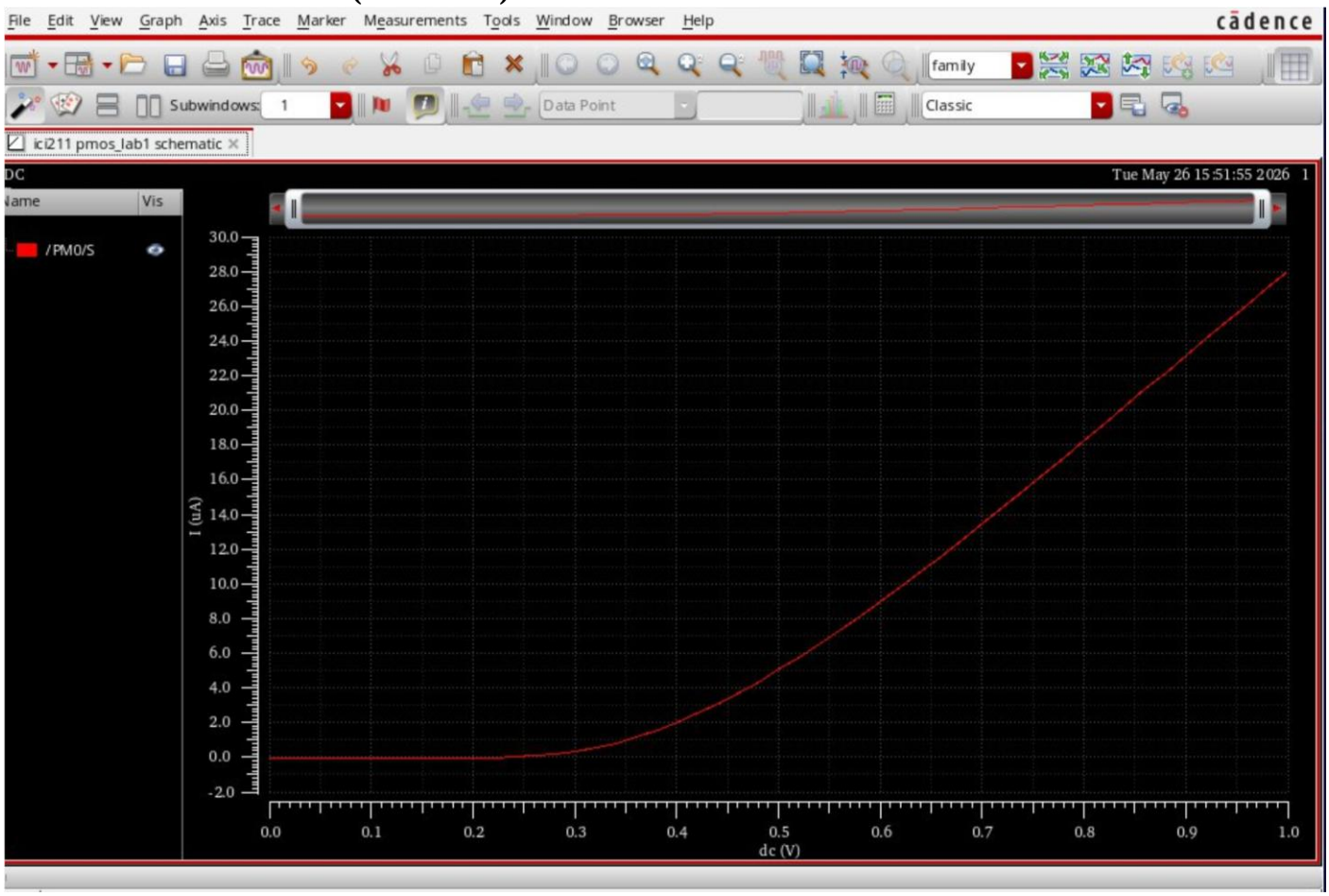
B.NMOS Characteristic Analysis

1.OutputCharacteristics(IDvsVDS)



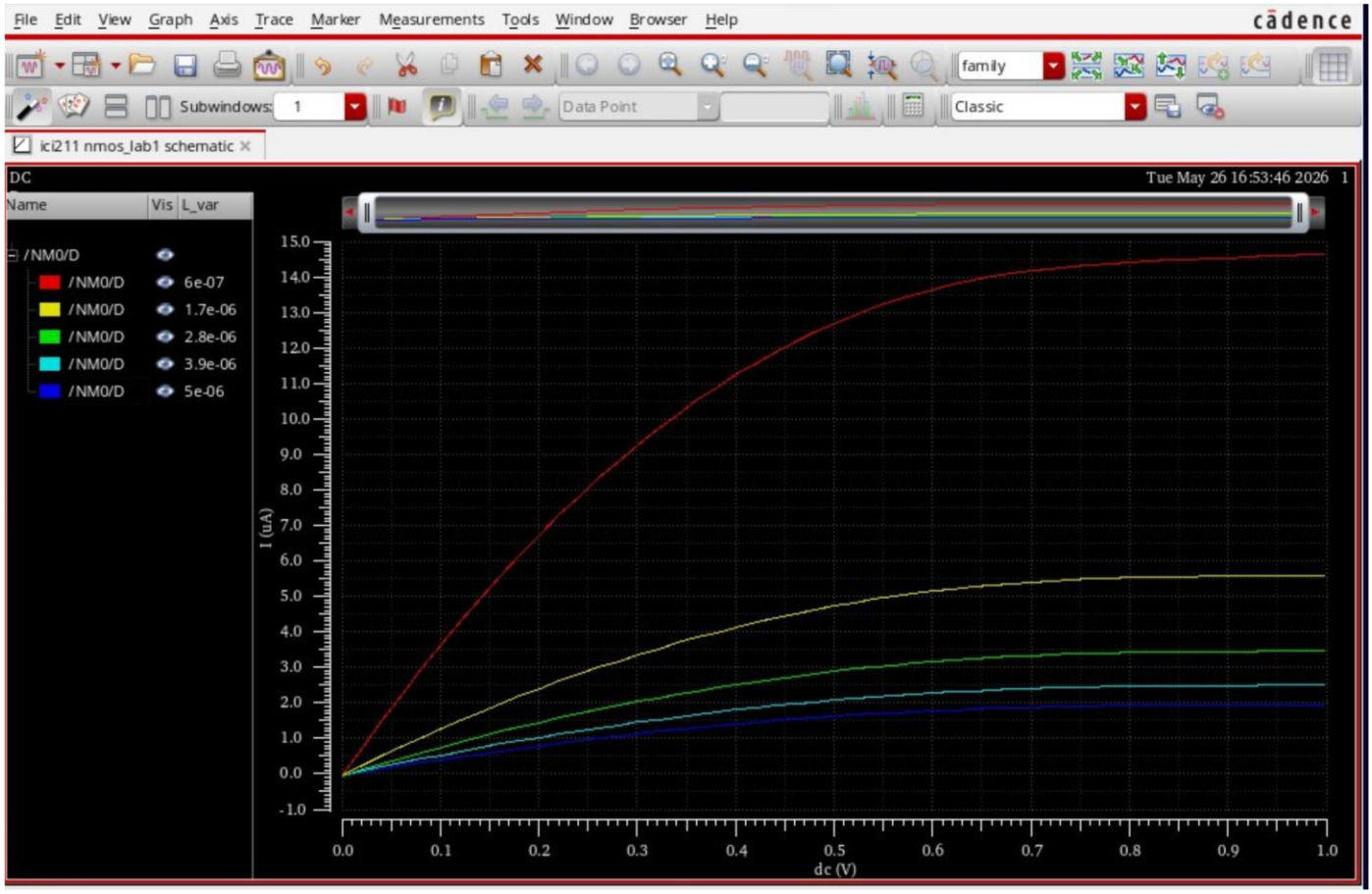
This simulation plot presents the output characteristics of an NMOS transistor, demonstrating the strong dependence of the drain current (I_D) on the gate-to-source bias (V_{GS}). As the V_{GS} parametric sweep increases, the inversion layer deepens, leading to a marked increase in I_D. The family of curves clearly depicts the transistor's operational transition from the linear (triode) region into the saturation region. Importantly, the saturation curves exhibit a noticeable positive slope rather than remaining perfectly flat. This behavior is a direct consequence of the Channel Length Modulation (CLM) effect, confirming that the NMOS device has a finite intrinsic output resistance (r_o). Accurately modeling this finite r_o is critical for the design of high-gain analog amplifier stages**.**

2.Transfer Characteristics (ID vs VGS)



This simulation plot presents the transfer characteristics ($\mathbf{I_D}$ vs. $\mathbf{V_{GS}}$) of the NMOS transistor, demonstrating the critical dependency of the drain current on the gate-to-source bias. At voltage levels below the 0.3V to 0.4V range, the device operates in the subthreshold (weak inversion) regime, where the channel is not fully formed and current flow is negligible. Once $\mathbf{V_{GS}}$ exceeds the threshold voltage ($\mathbf{V_{th}}$), the transistor enters the strong inversion region. This transition causes a rapid, non-linear increase in $\mathbf{I_D}$, consistent with the classic square-law model, ultimately reaching approximately 28 µA at 1.0V. From an analog design standpoint, the defining feature of this curve is its derivative or slope, which dictates the device's transconductance ($\mathbf{g_m}$). A steeper slope indicates a higher $\mathbf{g_m}$, a fundamental requirement for maximizing both the intrinsic voltage gain and the high-frequency performance of amplifier stages.

3.Effect of W/L Ratio:



This simulation graph demonstrates the parametric effect of varying the channel length (***L***) on the NMOS drain current (I_D). By sweeping ***L*** from 600 nm to 5 µm while maintaining a constant width (W), the device's aspect ratio ***W/L***) is effectively reduced, leading to a proportional decrease in its overall current-carrying capacity. Notably, while the minimum channel length (indicated by the red curve) provides the maximum current drive, it suffers from a pronounced positive slope in the saturation region due to severe Channel Length Modulation (CLM). Conversely, employing longer channel lengths (e.g., the blue curve) yields significantly flatter saturation characteristics. This intentional sizing strategy increases the intrinsic output resistance (r_o), representing a fundamental design trade-off necessary to maximize intrinsic voltage gain and enhance device matching in precision analog amplifiers.